

"Kestrel 1.5T: P0420 catalyst efficiency below threshold from degraded upstream

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Area: bulletins

1 Condition

Kestrel 1.5T vehicles may present with an illuminated MIL and stored DTC P0420 (Catalyst System Efficiency Below Threshold, Bank 1) without any accompanying driveability complaint. The code typically sets after extended highway operation once the monitor completes. Before condemning the catalytic converter, technicians must verify the upstream and downstream oxygen sensor switching behavior, as a lazy upstream sensor frequently triggers this code. See Kestrel Emissions Service Manual :: Catalyst Efficiency Faults for the full P0420 diagnostic tree.

2 Cause

In affected vehicles the root cause is a degraded upstream oxygen sensor, part number O2U-5100, whose slowed switching rate causes the PCM to misinterpret converter oxygen storage and falsely set P0420. The catalytic converter CAT-5500 is rarely the true cause and should only be replaced after the upstream sensor has been ruled out. Review oxygen sensor operation in Kestrel Emissions Service Manual :: Upstream and Downstream Oxygen Sensors and converter function in Kestrel Emissions Service Manual :: Catalytic Converter (Bank 1 and Bank 2) .

3 Repair Procedure

Replace the upstream oxygen sensor O2U-5100 following PROC-O2-SENSOR, taking care to apply anti-seize only to the threads and to route the harness clear of exhaust heat. Step-by-step removal and torque values are given in Kestrel Emissions Service Manual :: Oxygen Sensor Replacement Procedure . Background on the oxygen sensing subsystem is available in Kestrel Emissions Service Manual :: Oxygen Sensing . Do not replace CAT-5500 under this bulletin unless post-repair verification still indicates low catalyst efficiency.

4 Verification

After replacing the upstream sensor, clear P0420 and complete a drive cycle to allow the catalyst monitor to run to completion. Confirm the upstream sensor now switches rapidly and the downstream sensor remains relatively steady, indicating adequate oxygen storage. If P0420 returns after the monitor completes, proceed with full converter evaluation per Kestrel Emissions Service Manual :: Emissions System Diagnosis . Document the switching data captured during verification in the repair order.